Unit: Unit 2: Exponents Radicals
Topic: Radicals

Name: ____________
Date: ________

Advanced (Habanero)

Q1. Simplify $\sqrt{16}$
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10

Q2. Simplify $\sqrt[3]{27}$
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Q3. Convert to radical form: $x^{1/2}$
(A) $\sqrt{x}$
(B) $\sqrt[3]{x}$
(C) x^2
(D) x^3
(E) x^4

Q4. Simplify $\sqrt{25x^2}$
(A) $5x$
(B) $5\sqrt{x}$
(C) $\sqrt{5x}$
(D) $25x$
(E) $25\sqrt{x}$

Q5. Convert to rational exponent form: $\sqrt[3]{x^2}$
(A) $x^{2/3}$
(B) $x^{3/2}$
(C) $x^{1/2}$
(D) $x^{1/3}$
(E) $x^{2/5}$

Q6. Simplify $\sqrt{64x}$
(A) $8\sqrt{x}$
(B) $8x$
(C) $4\sqrt{2x}$
(D) $4x$
(E) $2\sqrt{2x}$

Q7. Convert to radical form: $y^{3/4}$
(A) $\sqrt[3]{y}$
(B) $\sqrt[4]{y^3}$
(C) $\sqrt{y}$
(D) y^3
(E) $y^{1/4}$

Q8. Simplify $\sqrt[3]{-8}$
(A) -2
(B) 2
(C) -1
(D) 1
(E) 0

Open Ended Questions (FRQ)

Q9. Simplify $\sqrt{48x^3}$. Show all steps.

Q10. Convert $x^{2/3}$ to radical form. Explain the reasoning behind your answer.

Chef's Tips

- Remind students to simplify radicals where possible
- Emphasize the importance of showing all steps in free response questions
- Encourage students to check their work carefully